



UNIFIED ERP — ANY INDUSTRY

Vendor Management

Vendor 360 · Payables · Aging · Risk · Performance-Based Sourcing

Domain 9 · Purchase — App 2 of 2

An industry-neutral supplier master. The same engine runs a textile mill, a medical distributor, a manufacturer or a services firm — you change only the vendor master.

Single-file · offline

14/14 self-tests

Wired to Procurement · Finance · Quality

What this is & what's inside

Vendor Management is the **one supplier truth** in the ERP. It answers four questions no spreadsheet answers together: **who is this supplier, what have we spent, what do we still owe, and can we rely on them?**

Procurement (App 1) creates the transactions. This app turns that history into **judgement** — a performance score from real receipts, a risk score that catches danger even when performance looks perfect, and a sourcing recommendation you can defend with numbers.

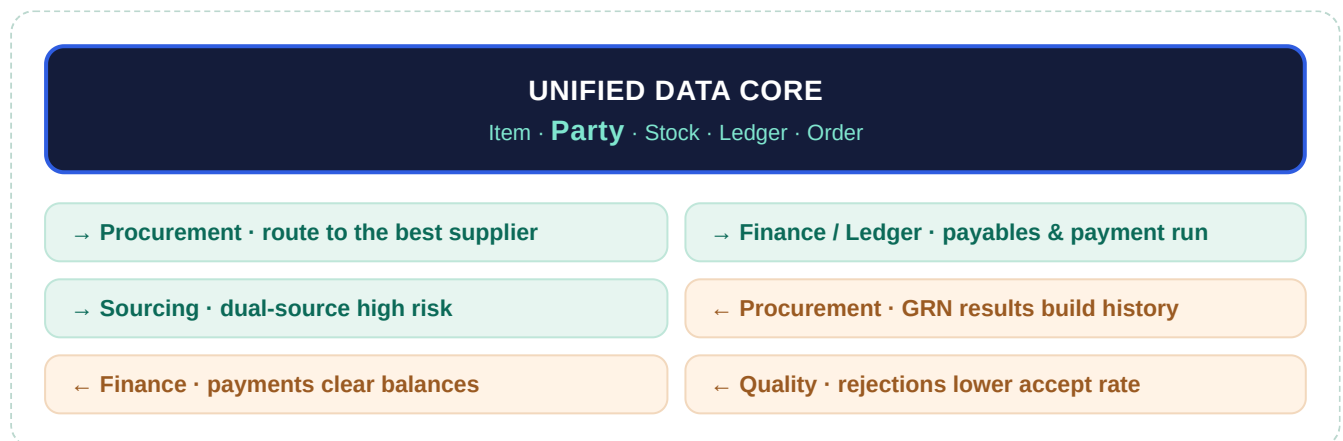
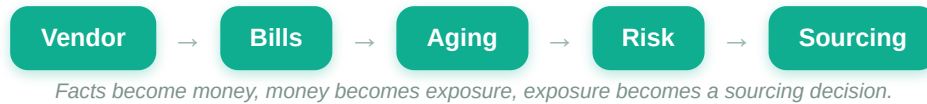
It is one self-contained HTML file. Double-click to open, runs **offline**, saves in the browser, and checks itself with **14 self-tests** on every launch.

Contents

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2. Where it sits — the Party entity
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Where Vendor Management sits

Every app in Medhava sits on **one shared Data Core**: Item/SKU, **Party**, Stock, Ledger/Voucher, Order. This app **owns the Party entity** for suppliers — so a vendor edited here is the same vendor Procurement raises a PO against and Finance pays.



Data model

Three entities, and everything numeric is **computed** — never re-keyed.

ENTITY	KEY FIELDS
Vendor	id · name · gstin · category · payment terms · location
Bill	id · vendor · date · due · amount · paid
Delivery	vendor · ordered · received · accepted · onTime — the performance history

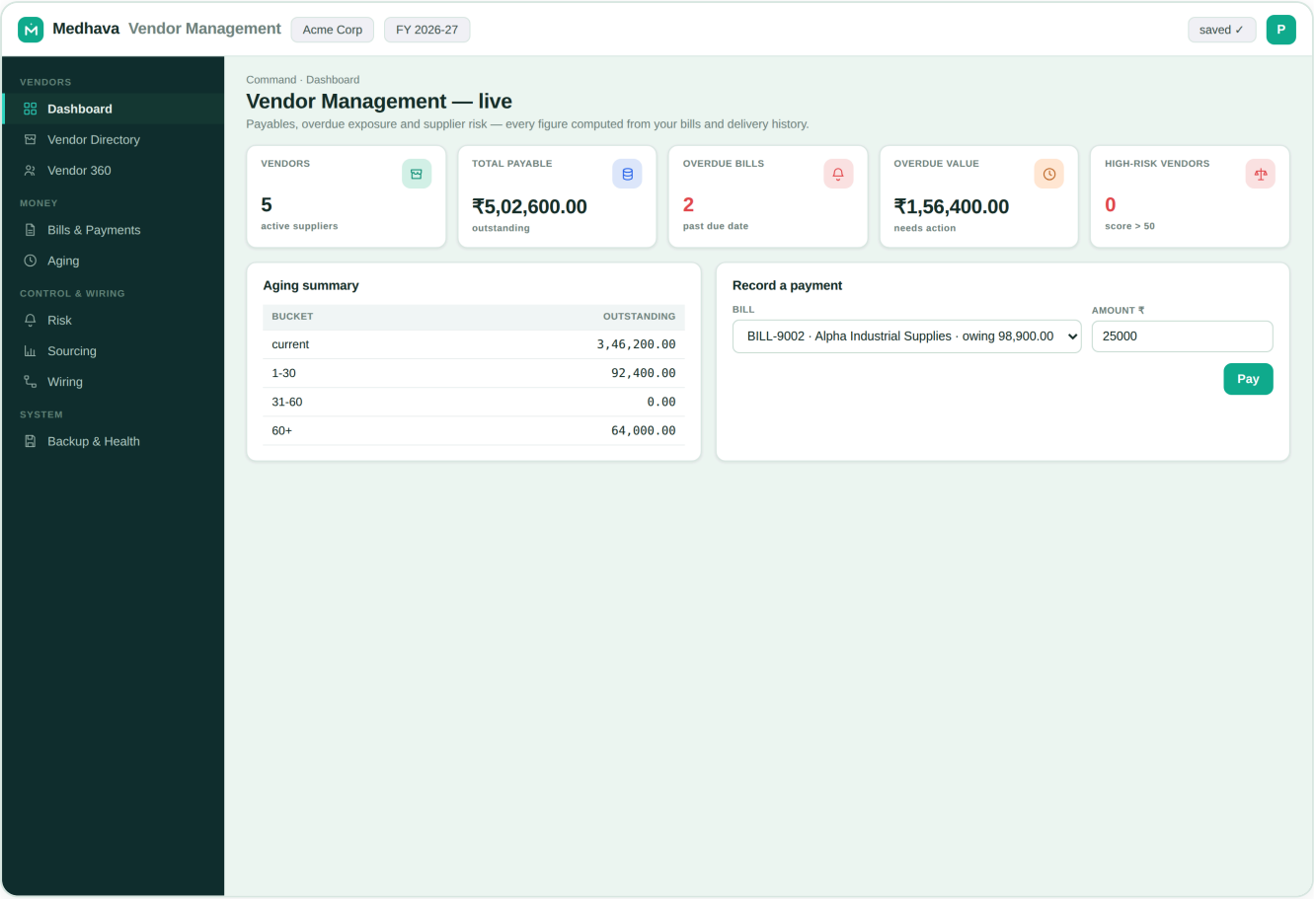
Nothing stores "outstanding", "risk" or "score". All three are derived on read, so they can never drift out of step with the underlying bills and receipts.

The demo dataset

FIGURE	VALUE
Total spend	₹8,00,300
Total payable	₹5,02,600
Overdue	2 bills · ₹1,56,400
Aging	current 3,46,200 · 1-30 92,400 · 31-60 0 · 60+ 64,000

Dashboard

The supplier position in one screen — what you owe, what is late, and how many suppliers are carrying real risk.



Payables, overdue exposure, high-risk count, aging summary and a payment form.

- **Overdue value** is the number that costs you relationships — it is computed, not tracked by hand.
- **High-risk suppliers** counts anything scoring above 50 on the risk model (page 13).
- Recording a payment updates the balance, the bucket and the risk score in one action.

Vendor Directory

Every supplier on one line: who they are, what share of your spend they hold, what you owe, and how they actually perform.

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FY 2026-27

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Vendors - Directory

Vendor directory

Every supplier with GSTIN, terms, spend share, payable and live performance score.

Suppliers

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ID	VENDOR	CATEGORY	GSTIN	TERMS	SPEND	SHARE	PAYABLE	SCORE	
V1	Alpha Industrial Supplies	Raw material	27AAAAA0001A1Z1	30 days	3,52,600.00	44%	98,900.00	99%	360 →
V2	Beta Components Ltd	Components	29BBBBB0002B1Z2	15 days	1,50,200.00	19%	1,50,200.00	64%	360 →
V3	Gamma Materials Co	Raw material	24GGGGG0003C1Z3	45 days	1,89,500.00	24%	1,89,500.00	100%	360 →
V4	Delta Trading	Consumables	07DDDDD0004D1Z4	30 days	64,000.00	8%	64,000.00	97%	360 →
V5	Epsilon Enterprises	Packaging	19EEEEEE0005E1Z5	COD	44,000.00	5%	0.00	98%	360 →

Spend, share %, payable and a colour-coded performance score per vendor.

- ▶ Performance badge: **green** ≥ 90%, **amber** 70–89%, **red** < 70%.
- ▶ **Share %** is the early-warning column — it is what turns a good supplier into a risky one.
- ▶ "360 →" opens the full record for that supplier.

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Vendor 360

One supplier, completely. Spend and share, live payable, delivery performance broken into its three parts, risk band, full profile and every bill.

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Vendors - 360

Vendor 360 — Alpha Industrial Supplies

Everything about one supplier on one screen: terms, spend, bills, performance and risk.

Choose vendor

VENDOR

Alpha Industrial Supplies

Open 360

TOTAL SPEND

₹3,52,600.00

44% of all spend

PAYABLE NOW

₹98,900.00

outstanding

PERFORMANCE

99%

2 deliveries

RISK

27

medium

Profile

Category	Raw material
GSTIN	27AAAAA0001A1Z1
Payment terms	30 days
Location	Pune, MH
On-time	100%
Quality (accept rate)	98%
Fill rate	100%

Bills

BILL	DUE	AMOUNT	OWING	STATUS
BILL - 9001	2026-07-05	2,03,700.00	0.00	paid
BILL - 9002	2026-07-28	1,48,900.00	98,900.00	current

Spend, payable, performance and risk, with profile and bill history.

- ▶ Alpha Industrial Supplies: ₹3,52,600 spend — 44% of everything you buy.
- ▶ On-time 100%, quality 98%, fill 100% → performance 99%.
- ▶ Yet risk is 27 (medium) — because of that 44%. Excellence and exposure are different things.

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Bills & Payments

The payables ledger. Every bill with amount, paid, still owing, and which aging bucket it has fallen into.

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Money · Bills & Payments

Bills & payments

Every supplier bill with what is still owed. Paying reduces the balance live.

Bills

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BILL	VENDOR	DATE	DUE	AMOUNT	PAID	OWING	STATUS
BILL-9001	Alpha Industrial Supplies	2026-06-05	2026-07-05	2,03,700.00	2,03,700.00	0.00	paid
BILL-9002	Alpha Industrial Supplies	2026-06-28	2026-07-28	1,48,900.00	50,000.00	98,900.00	current
BILL-9003	Beta Components Ltd	2026-06-10	2026-06-25	92,400.00	0.00	92,400.00	1-30
BILL-9004	Gamma Materials Co	2026-07-02	2026-08-16	1,89,500.00	0.00	1,89,500.00	current
BILL-9005	Delta Trading	2026-04-20	2026-05-10	64,000.00	0.00	64,000.00	60+
BILL-9006	Epsilon Enterprises	2026-07-10	2026-07-10	44,000.00	44,000.00	0.00	paid
BILL-9007	Beta Components Ltd	2026-07-15	2026-07-30	57,800.00	0.00	57,800.00	current

Add a bill

VENDOR

Alpha Industrial Supplies

AMOUNT ₹

50000

DUE DATE

2026-08-25

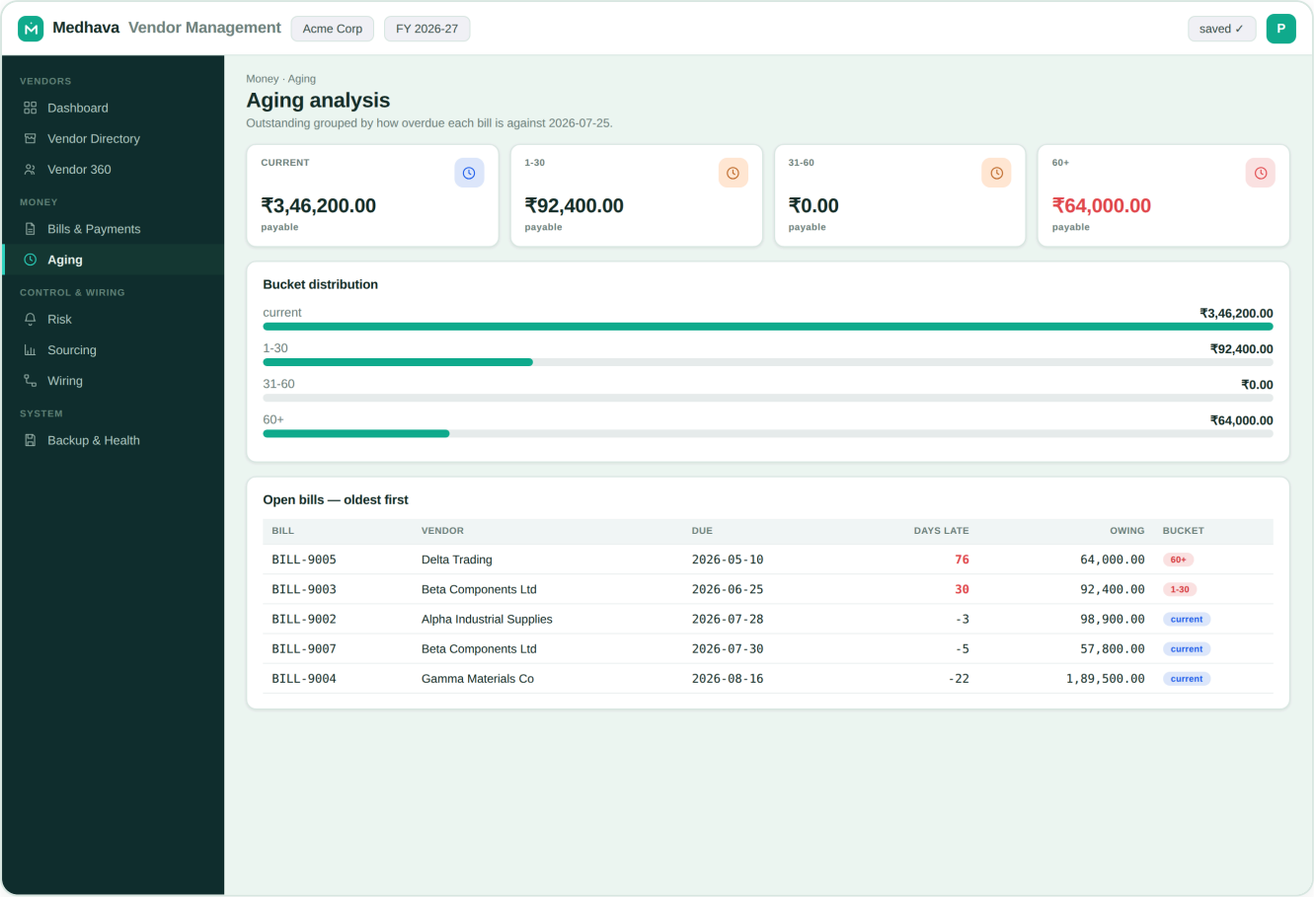
Add bill

Amount / paid / owing / bucket for each bill, plus an add-bill form.

- ▶ **Owing = amount – paid**, floored at zero — you cannot overpay a bill into a negative balance.
- ▶ Paying more than is owed is clamped to the outstanding amount.
- ▶ Status is derived from the due date against today, never typed in.

Aging

How late is the money? Every open bill is bucketed against today, so you can see at a glance what must clear this week versus what has become a problem.

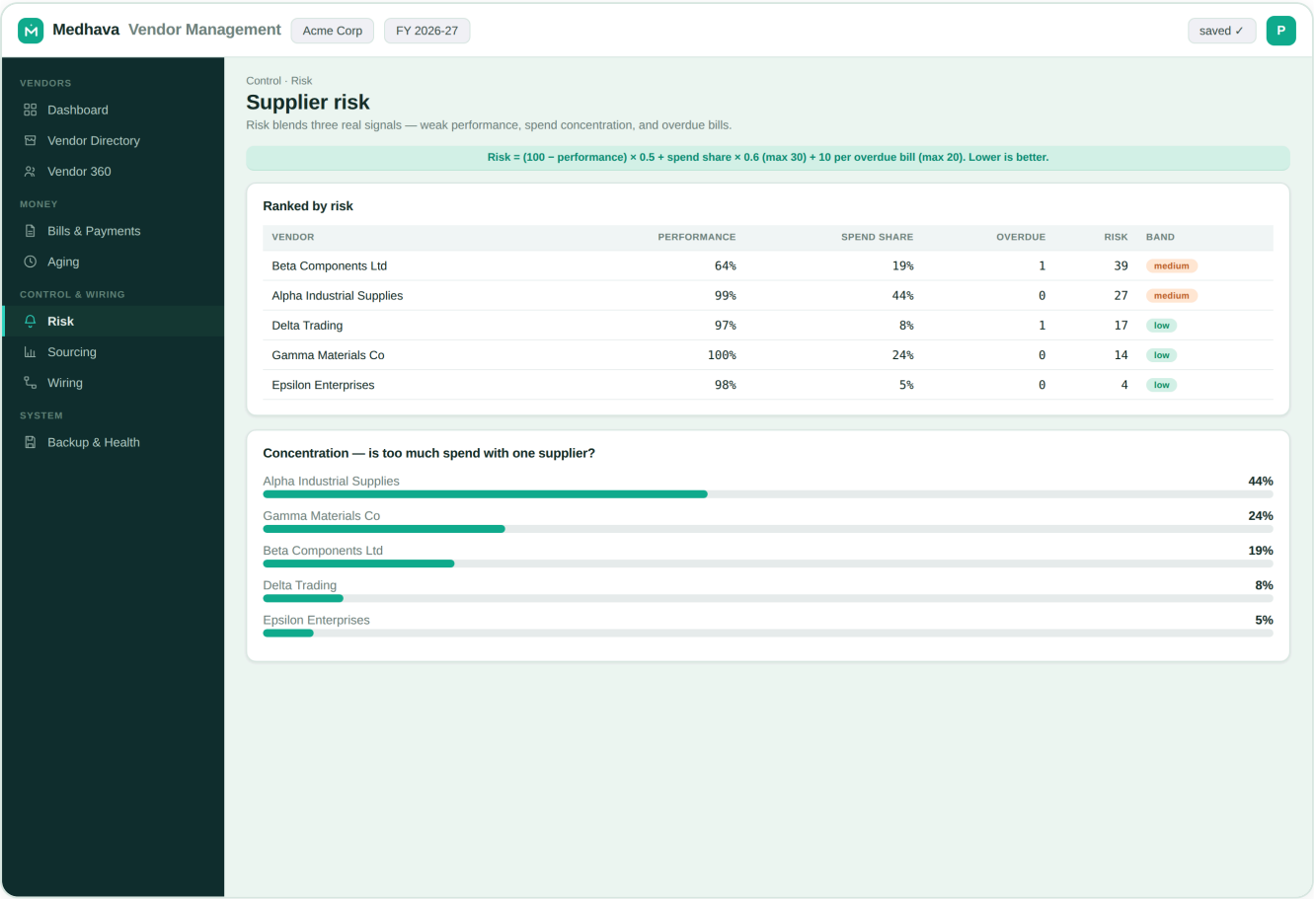


Bucket KPIs, a distribution chart, and open bills sorted oldest-first.

- ▶ 60+ · ₹64,000 — Delta Trading, 76 days late. That is the one to call today.
- ▶ 1-30 · ₹92,400 — Beta Components Ltd, 30 days late.
- ▶ Buckets always sum exactly to total payable — self-test #4 enforces it.

Risk — the control

The screen that earns the app its place. Risk is not a feeling; it is three measurable signals combined, and it deliberately catches danger that a performance score alone would hide.



Vendors ranked by risk, with the formula stated openly and a concentration chart.

- ▶ **Beta Components Ltd — 39 (medium):** 64% performance and one overdue bill.
- ▶ **Alpha Industrial Supplies — 27 (medium) at 99% performance.** Nothing is wrong with them; you simply depend on them too much.
- ▶ The formula is printed on the screen, so no one has to trust a black box.

Sourcing

The output of everything else: for each category, the supplier the evidence says you should buy from — with a watchlist that states plainly why anyone is flagged.

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Control - Sourcing

Performance-based sourcing

For each category the engine names the best-performing supplier — sourcing follows evidence, not habit.

Preferred vendor by category

CATEGORY	PREFERRED VENDOR	SCORE	RISK
Raw material	Gamma Materials Co	100%	low
Components	Beta Components Ltd	64%	medium
Consumables	Delta Trading	97%	low
Packaging	Epsilon Enterprises	98%	low

Best overall

Gamma Materials Co is your strongest supplier at 100% — on-time 100%, quality 99%, fill 100%. Route new orders here first.

Watchlist

VENDOR	WHY
Alpha Industrial Supplies	concentration 44%
Beta Components Ltd	performance 64% · 1 overdue

Preferred vendor per category, best overall, and a reasoned watchlist.

- ▶ **Gamma Materials Co at 100%** is the strongest supplier overall — new orders route there first.
- ▶ Every watchlist entry names its cause: *performance 64% · concentration 44% · 1 overdue*.
- ▶ This turns "we've always bought from them" into a decision you can defend.

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Wiring — one record, every module

The vendor master is a Data Core entity, so it is never a private list. Every module that touches a supplier reads and writes the same record.

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Wiring - Integration

How Vendor Management wires into the rest of the ERP

The vendor master is a Data Core entity — Party. Every module that touches a supplier reads and writes it here.

Shared Data Core: Item/SKU · Party (vendor) · Stock · Ledger/Voucher · Order — every module reads and writes these.

Outbound data flows (this app → others)

TRIGGER (HERE)	FLows TO	WHAT MOVES
Vendor created / updated	Master Data (Party)	One supplier identity every module reads — Procurement, Finance, Quality
Performance score	Procurement	RFQs and POs route to the best-performing supplier in that category
Payable balance	Finance / Ledger	Accounts-payable position and the payment run
Aging buckets	Treasury / Cash-flow	What must be paid this week vs this month
Risk band	Sourcing & Management	High-risk suppliers trigger dual-sourcing and a review
Spend concentration	Board / BI	Single-supplier dependency flagged before it becomes a failure

Live example — Beta Components Ltd

1

Two late deliveries drop on-time to 0%.

2

→ Risk: score rises to 39 (medium) — overdue bills add to it.

3

→ Sourcing: this vendor loses preferred status in its category.

4

→ Procurement: the next RFQ routes to the higher-scoring supplier instead.

5

→ Finance: ₹1,50,200.00 payable, 1 bill(s) overdue, flagged for the payment run.

Inbound (others → Vendor Management)

FROM	WHAT
Procurement	GRN results (received / accepted / rejected, on-time) build the performance history
Finance	Payments post against bills and clear the outstanding balance
Quality	Rejections and debit notes lower the accept rate
Master Data	GSTIN, payment terms, tax and category definitions

Outbound and inbound flows, plus the live cascade from a late delivery.

- ▶ A late delivery → on-time drops → risk rises → preferred status lost → the next RFQ routes elsewhere → Finance flags the overdue bill.
- ▶ That is five modules reacting to one fact, with no one re-typing anything.

Formulas & rules

```
outstanding(bill) = r2(max(0, amount - paid))
daysLate(bill)   = days(due → TODAY)          TODAY = 2026-07-25 (fixed, so aging is testable)
bucket(bill)     = paid | current | 1-30 | 31-60 | 60+

payable(vendor?) = r2(Σ outstanding)
spend(vendor)    = r2(Σ amount)                sharePct = round(spend / totalSpend × 100)

PERFORMANCE (null when no delivery history)
  onTime% = round(onTimeDeliveries / totalDeliveries × 100)
  quality% = round(Σaccepted / Σreceived × 100)          // accept rate at inspection
  fill%    = round(Σreceived / Σordered × 100)
  score    = round(mean of available metrics)

RISK 0 best ... 100 worst
  performance = 25 if no history, else round((100 - score) × 0.5)    max 50
  concentration = min(30, round(sharePct × 0.6))                    max 30
  discipline    = min(20, overdueBillCount × 10)                     max 20
  band: ≤25 low · ≤50 medium · >50 high

PREFERRED VENDOR = highest score among rated vendors (per category, and overall)
```

Everything above is derived at read time. The database stores vendors, bills and receipts; the judgements are recomputed on every render.

The risk model explained

Most vendor scorecards measure only **performance**. That misses the two ways suppliers actually hurt you.

SIGNAL	MAX	WHY IT BELONGS
Weak performance	50	Late, short or rejected deliveries stop your production line. The largest single weight.
Spend concentration	30	If one supplier holds 44% of your buying, their bad quarter becomes your bad quarter — however well they perform today.
Payment indiscipline	20	Bills you have let run overdue damage the relationship and your priority in their queue. Risk you created.

The case that proves the model: Alpha Industrial Supplies scores **99% on performance** — near-perfect. A conventional scorecard would rate them your safest supplier. Medhava rates them **medium risk (27)**, because 44% of everything you buy comes from them. That is not a criticism of the supplier; it is a fact about *your* exposure, and it is exactly the fact worth knowing before a disruption, not after.

Self-tests — all 14 pass

These run on every launch and appear on **Backup & Health**. They pin the exact figures this document quotes.

#	CHECK	EXPECTED	RESULT
1	outstanding = amount – paid	₹98,900	✓ pass
2	fully paid bill is bucket "paid"	BILL-9001	✓ pass
3	total payable = Σ outstanding	₹5,02,600	✓ pass
4	bucket totals sum to total payable	= 5,02,600	✓ pass
5	BILL-9005 is 60+ days overdue	76 days late	✓ pass
6	Alpha Industrial Supplies on-time = 100%	2 of 2	✓ pass
7	Alpha Industrial Supplies quality = 98%	156 / 160 received	✓ pass
8	Beta Components Ltd on-time = 0%	0 of 2	✓ pass
9	Delta Trading fill rate = 90%	180 of 200	✓ pass
10	spend shares sum to ~100%	within rounding	✓ pass
11	risk bounded 0..100	every vendor	✓ pass
12	late + overdue is riskier than clean	39 > 14	✓ pass
13	preferred vendor is highest-scoring	Gamma Materials Co @ 100%	✓ pass
14	paying reduces payable exactly	by the amount paid	✓ pass

Tests run against a deep copy of the database, never the live one. A self-test that paid a bill would silently alter your books on every launch — so the harness clones first. Reload the app and every figure is identical.

Data in / out & hosted API

The single file is the **local edition**. The hosted edition runs the identical engine against a backend, so performance history is written automatically instead of seeded.

BOUNDARY	LOCAL (THIS FILE)	HOSTED EDITION
Storage	Browser localStorage + JSON backup	Postgres with row-level security per company
Performance history	Seeded deliveries	Written automatically by every GRN in Procurement
Payments	Recorded in-app	Posted from Finance / Ledger and bank reconciliation
Sourcing	Recommendation shown	Preferred vendor auto-applied to new RFQs
3rd-party auth	—	Revocable scoped API keys in an encrypted vault — never passwords

API surface (hosted): /vendor · /vendor/{id}/360 · /bill · /payment · /aging · /vendor/risk · /sourcing/preferred — each emits an event other modules subscribe to.

Honest limits & roadmap

- ▶ **Local-first, single browser.** Cross-module posting is real in the hosted edition, illustrated here.
- ▶ **Performance history is seeded** in this build; hosted, it is a by-product of goods receipt.
- ▶ **Not in this file:** multi-currency, vendor self-service portal, contract & rate-card expiry, TDS on vendor payments, credit-limit blocking — all hosted-tier.
- ▶ **Risk weights are a starting point.** They are printed on screen precisely so you can argue with them and tune them to your business.
- ▶ **Security:** real connections use scoped, revocable keys in an encrypted vault. Account passwords are never used or stored.

Domain 9 · Purchase

✓ Procurement

·

✓ Vendor Management

Domain 9 complete. Next: Domain 7 · Inventory, then Domain 5 · Warehouse.

How to run & accept

1. **Open** `Vendor Management.html` by double-clicking — any modern browser, offline.
2. **Backup & Health** → confirm **14/14 pass**. Refresh; every figure is unchanged.
3. **Aging** → ₹64,000 in **60+** (Delta Trading, 76 days late).
4. **Risk** → Beta Components Ltd 39 (medium); **Alpha Industrial Supplies 27 (medium) at 99% performance**.
5. **Sourcing** → best overall **Gamma Materials Co at 100%**, with a reasoned watchlist.
6. **Vendor 360** → Alpha Industrial Supplies: ₹3,52,600 spend, 44% share, on-time 100%, quality 98%.
7. **Wiring** → the Beta Components Ltd cascade across five modules.

Accepted when: it opens offline, all 14 tests are green, no console errors, figures survive a refresh, and the numbers match your own records.

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